

Computational Method in Chemical Engineering T2730C2

Textbooks:

- Lorens T. Biegler, E. Ignacio Grossmann, Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International.
- Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation, 2nd Edition, Wiley.
 - Robin Smith, Chemical Process: Design and Integration, Wiley.
- James M. Douglas, Conceptual Design of Chemical Processes, McGraw Hill International, 1988.

Unit 2 – Reactor Design and Reactor Network Synthesis

- Objectives
- Reactor Models
- Reactor Design for complex configurations
- Reactor Network Design using the Attainable Region

Objectives

Objectives:

1. Learn types of reactor models available in the simulators and their use in process simulation. (ASPEN Chemical Reactors and HYSYS Chemical Reactors).
2. Be able to design a system for heat transfer in association with the reactor, to sustain an exothermic or endothermic reaction at its desired temperature level, and study the design using simulation.
3. Be able to determine if a reactor network should be considered and, if so, design it using the concept of the attainable region.

Objectives of reactor design

- The design of the reactor section of the process flow diagram addresses the need to eliminate differences in molecular type.
- **It is desired to ensure sufficient yield and selectivity of the required product species, by appropriate selection of a single reactor or net work of reactors.**
- **Care is required in heat exchange in endothermic and exothermic reactors, to sustain stable operation at the desired temperature level.**
- Types of reactor models available in the flowsheet simulators, discussing also the applicability of the reactor types to model particular process circumstances.
- Arrangements for heat transfer in exothermic and endothermic reactors, with examples from the processing industry.
- Attainable region analysis as a tool for the optimal design of reactor networks.

Reactor models

- Chemical reactors, particularly for continuous processes, are custom designed to involve **factors**:
 1. multiple phases (e.g., vapor, liquid, reacting solid, and solid catalyst),
 2. different geometries (e.g., stirred tanks, tubular flows, converging and diverging nozzles, spiral flows, and membrane transport),
 3. various regimes of momentum, heat, and mass transfer (e.g., viscous flow, turbulent flow, conduction, radiation, diffusion, and dispersion).
- There are so many configurations, involving different combinations of these attributes, that attempt to develop generalized reactor models have met with limited success.

Reactor models

- Most of the process simulators provide **four kinds of reactor models**, including:
 1. A **stoichiometric model** that permits the specification of reactant conversions and extents of reaction for one or more specified reactions
 2. A **model for multiple phases** (vapor, liquid, and solid) **in chemical equilibrium**, where the approach to equilibrium for individual reactions can be specified;
 3. A kinetic model for a **continuous-stirred-tank reactor (CSTR)** that assumes perfect mixing of homogeneous phases (liquid or vapor)
 4. A kinetic model for a **plug-flow tubular reactor (PFTR or PFR)**, for homogeneous phases (liquid or vapor) and assuming no backmixing (dispersion).

Reactor models

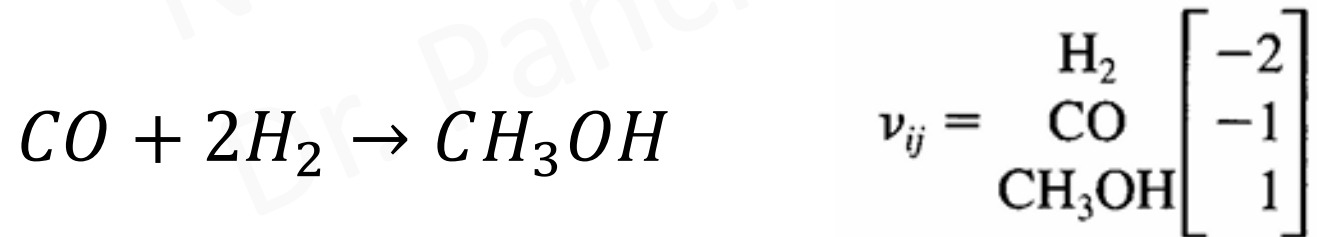
- These **ideal models** are used in the **early stages of process synthesis**, when the details of the reactor designs are less important, but reactor effluents and heat duties are needed.
- The ideal reactor models are **replaced by custom-made models** as the details gain significance.
- For this purpose, all of the **flowsheet simulators** provide facilities for the insertion of user-generated models.
- These are refined often as the design proceeds and as **reactor data from the laboratory or pilot plant** are regressed, with some of the simulators providing facilities for estimating the parameters of kinetic models by nonlinear regression.

Reaction Stoichiometry

- For most of the reactor models in the flowsheet simulators, it is necessary to provide R chemical reactions involving C chemical species:

$$\sum_{j=1}^C \nu_{ij} A_j = 0, i = 1, \dots, R$$

- Where A_j is the chemical formula for species j and ν_{ij} is the **stoichiometric coefficient** for species j in reaction i (negative for reactants, positive for products).



Extent of Reaction

- For a single reaction, in the stoichiometric reactor models, one specifies **the fractional conversion**, X_k of key reactant k

$$X_k = \frac{n_{k,in} - n_{k,out}}{n_{k,in}},$$

- where $n_{k,in}$ and $n_{k,out}$ are the moles of species k entering and leaving the reactor and $0 \leq X_k \leq 1$
- The **extent (number of moles extent)** of reaction i ,

$$\xi_i = \frac{\Delta n_{ij}}{\nu_{ij}}, j = 1, \dots, C$$

Extent of Reaction

Component material balance:

- The molar flow rates of the components in the reactor effluent, $n_{j,out}$ are computed from the component molar flow rates in the reactor feed, $n_{j,in}$ by a component material balance equation that is consistent with the reaction stoichiometry.
- If a specification of the fractional conversion of the key component, k

$$n_{j,out} = n_{j,in} - n_{k,in} X_k \left(\frac{\nu_j}{\nu_k} \right), j = 1, \dots, C$$

$$n_{j,out} = n_{j,in} + \xi \nu_j, \quad j = 1, \dots, C$$

Extent of Reaction

- **Ex.** For the conversion of CO and H₂ to CH₃OH, assuming an initial feed of 100 kmol/hr CO and 600 kmol/hr H₂ and 70% conversion of CO (the key component).
- The molar flow rates of the three components in the reactor effluent are

$$n_{j,out} = n_{j,in} - n_{k,in} X_k \left(\frac{v_j}{v_k} \right), j = 1, \dots, C$$

$$n_{H_2,out} = 600 - 100(0.7)(-2/-1) = 460 \text{ kmol/hr}$$

$$n_{CO,out} = 100 - 100(0.7)(-1/-1) = 30 \text{ kmol/hr}$$

$$n_{CH_3OH,out} = 0 - 100(0.7)(1/-1) = 70 \text{ kmol/hr}$$

- Mole fraction of methanol in reactor effluent is $y_{CH_3OH} = 70/(460 + 30 + 70) = 0.125$.
- If extent of reaction is specified as 70 kmol/hr $n_{j,out} = n_{j,in} + \xi v_j, \quad j = 1, \dots, C$
- $n_{H_2,out} = 600 + (70)(-2) = 460 \text{ kmol/hr}$ $n_{CO,out} = 100 + (70)(-1) = 30 \text{ kmol/hr}$
- $n_{CH_3OH} = 0 + (1)(70) = 70 \text{ kmol/hr}$

Extent of Reaction

- For multiple reactions, the reactions must be specified as series or parallel.
- The former is equivalent to having reactors in series with the feed to each reactor (except the first) being the product from the previous reactor.
- Each reaction can have a different key reactant.
- For parallel reactions, it is preferable to specify the extent of reaction for each reaction, which results in

$$n_{j,out} = n_{j,in} + \sum_{i=1}^R \xi_i \nu_{ij}, \quad j = 1, \dots, C$$

Equilibrium

- A **reaction** is characterized by two important thermodynamic quantities, namely the **heat of reaction and the Gibbs (free) energy of reaction**.
- These two quantities are **functions of temperature and pressure**.
- **Thermodynamic data** are widely available in simulators and elsewhere, for more than a thousand components, for the calculation of these two quantities under standard state conditions (at a reference temperature of 25°C and 1 bar with all components in a designated phase, usually as an ideal gas).
- The effect of temperature on the heat of reaction depends on the **heat capacities of the reactants and products and the effect of temperature on those heat capacities**.
- *For many reactions, the effect of temperature on the heat of reaction is relatively small however the effect on Gibbs free energy is substantial.*

Equilibrium

- Many reactions of industrial importance are limited by **chemical equilibrium**, with partial conversion of the limiting reactant and, with the rate of the reverse reaction equal to the rate of the forward reaction.
- For a specified feed composition and final temperature and pressure, **the product composition at chemical equilibrium can be computed by either of two methods:**
 1. **Chemical equilibrium constants (K-values) computed from the Gibbs energy of reaction combined with material balance equations** for a set of independent reactions; applicable when stoichiometry can be specified for all reactions being considered
 2. **Minimization of the Gibbs energy of the reacting system**; requires only a list of the possible products

Equilibrium – 1st method

- Chemical equilibrium constant, K , is computed for each independent stoichiometric reaction in the set, using the equation, where a_i is the component activity.

$$K = \frac{a_R^r a_S^s \cdots}{a_A^a a_B^b \cdots} = \exp \left(- \frac{\Delta G_{\text{rxn}}^\circ}{RT} \right)$$

- For a gas solution, the activity is given by, where $\bar{\Phi}_i$ is the fugacity coefficient of component i in the gas mixture, $\bar{\Phi}_i$ equal to 1.0 for an ideal gas solution, and P_i is the partial pressure.

$$a_i = \bar{\Phi}_i y_i P = \bar{\Phi}_i P_i$$

$$K = \frac{y_R^r y_S^s \cdots}{y_A^a y_B^b \cdots} \left(\frac{\bar{\Phi}_R^r \bar{\Phi}_S^s \cdots}{\bar{\Phi}_A^a \bar{\Phi}_B^b \cdots} \right) P^{r+s+\cdots-a-b-\cdots} = \frac{y_R^r y_S^s \cdots}{y_A^a y_B^b \cdots} K_{\bar{\Phi}} P^{r+s+\cdots-a-b-\cdots} = \frac{P_R^r P_S^s \cdots}{P_A^a P_B^b \cdots} K_{\bar{\Phi}}$$

where $K_{\bar{\Phi}} = 1.0$ for low to moderate pressures.

Equilibrium – 1st method

- For a liquid solution, the activity is given as, where γ_i is the activity coefficient of component i in the liquid mixture and γ_i is equal to 1.0 for an ideal liquid solution, $\bar{V}_i$ is the partial molar volume of component i , and P_i^s is the vapor pressure of component i .

$$a_i = x_i \gamma_i \exp \left[\frac{\bar{V}_i}{RT} (P - P_i^s) \right]$$

- γ_i is a function of T , P , and composition.
- For ideal liquid solutions at low to moderate pressures, $\gamma_i = 1.0$, so that the activity is equal to the mole fraction.**

$$K = \frac{x_R^r x_S^s \cdots}{x_A^a x_B^b \cdots} \left(\frac{\gamma_R^r \gamma_S^s \cdots}{\gamma_A^a \gamma_B^b \cdots} \right) = \frac{x_R^r x_S^s \cdots}{x_A^a x_B^b \cdots} K_\gamma$$

- where $K_\gamma = 1.0$ for ideal liquid solutions at low to moderate pressures.

Equilibrium – 1st method

- Chemical thermodynamics present charts of $\log_{10} K$ as a function of temperature for many chemical reactions.
- The van't Hoff equation relates K to temperature by $\left(\frac{d \ln K}{dT} \right)_P = \frac{\Delta H_{\text{rxn}}^\circ}{RT^2}$
- If the heat of reaction is assumed independent of temperature over a particular range of temperature, integration and conversion to $\log_{10}$ form gives the approximate correlating equation, where T is the absolute temperature.

$$\log_{10} K = A + B/T$$

Equilibrium – 1st method

Typically, the methanol synthesis reaction is catalyzed by copper-zinc oxide, at a pressure of 100 bar and a temperature of 300°C. A large excess of hydrogen is used to help absorb the relatively high heat of reaction. At these conditions, $K = 0.0002202$ and $K_{\phi} = 0.61$. Therefore,

$$\frac{(y_{\text{CH}_3\text{OH}})}{(y_{\text{CO}})(y_{\text{H}_2})^2} = \frac{K}{K_{\phi}} P^2 = \frac{0.0002202}{0.61} (100)^2 = 3.61 \quad (6.17)$$

If X is the equilibrium fractional conversion of the limiting reactant, CO, then using the same initial feed composition as before and the stoichiometry for the reaction, the equilibrium mole fractions are

$$y_{\text{CH}_3\text{OH}} = \frac{100 X}{700 - 200 X}$$

$$y_{\text{CO}} = \frac{100 - 100 X}{700 - 200 X}$$

$$y_{\text{H}_2} = \frac{600 - 200 X}{700 - 200 X}$$

Combining the above four equations to give a nonlinear equation in X and solving gives $X = 0.7087$.

Equilibrium – 2nd method

- The second method for computing chemical equilibrium is to apply the criterion that the total Gibbs energy, G , is a minimum at constant temperature and pressure.
- Two advantages of this second method are: (1) the avoidance of having to formulate stoichiometric equations (only the possible products need to be specified), and (2) the ease of formulation for multiple phases and simultaneous phase equilibrium.
- For a single phase, the total Gibbs energy at a specified T and P is given by $G = \sum_{i=1}^C N_i \bar{G}_i$
- where N_i is the mole number of component i and $\bar{G}_i$ is the partial molar Gibbs energy of component i in the equilibrium mixture.
- **The Gibbs energy is minimized with respect to N_i which are constrained by atom balances.**
- The method is readily extended to multiple phases.

Kinetics

- Fractional conversion and equilibrium reactor models are useful in the early stages of process design when conducting material and energy balance studies.
- **Eventually reactor systems must be configured and sized, which requires knowledge of reaction kinetics obtained by conducting laboratory experiments.**
- For homogeneous non catalytic reactions, power-law expressions are commonly used for regression of laboratory kinetic data.
- Expressions are not always based on the stoichiometric equation as several elementary reaction steps may be involved but of the one which may control overall reaction rate.
- The general **power-law kinetic equation** is
$$-r_j = -\frac{dC_j}{dt} = k \prod_{i=1}^C C_i^{\alpha_i}$$
- where $-r_j$ is the rate of disappearance of component j (in mol/time-volume), C_i is the concentration of component i (mol/volume), t is time, k is the reaction rate coefficient, α_i is the order of reaction with respect to component i, and C is the number of components.

Kinetics

- For gas-phase reactions, the partial pressure, P_i is sometimes used in place of the concentration C_i in the kinetic equation.
- **The reaction rate coefficient** is a function of temperature as given by the empirical **Arrhenius equation**, where k_o is the pre-exponential factor, and E is the activation energy.

$$k = k_o \exp(-E/RT)$$

- For reactions that are catalyzed by solid porous catalyst particles, the sequence of elementary steps may include adsorption on the **catalyst surface** of one or more reactants and/or desorption of one or more products.
- **Langmuir-Hinshelwood (LH) kinetic equation** is often found to fit the experimental kinetic data more accurately than the power law expression.

Kinetics

- The LH formulation is characterized by a denominator term that includes concentrations of certain reactants and/or products that are strongly adsorbed on the catalyst.
- The LH equation may also include a prefix η called **an overall effectiveness factor that accounts for mass and heat transfer resistances**, both external and internal, to the catalyst particles.
- Ex. laboratory kinetic data for the air-oxidation of SO_2 to SO_3 are fitted well by the following LH equation, where K is the chemical equilibrium constant and K_1 and K_2 are adsorption equilibrium constants.

$$-r_{\text{SO}_2} = \frac{\eta k \left(P_{\text{SO}_2} P_{\text{O}_2} - \frac{P_{\text{SO}_3}^2}{K^2 P_{\text{SO}_2}} \right)}{[1 + K_1 P_{\text{SO}_2}^{1/2} + K_2 P_{\text{SO}_3}^{1/2}]}$$

CSTR

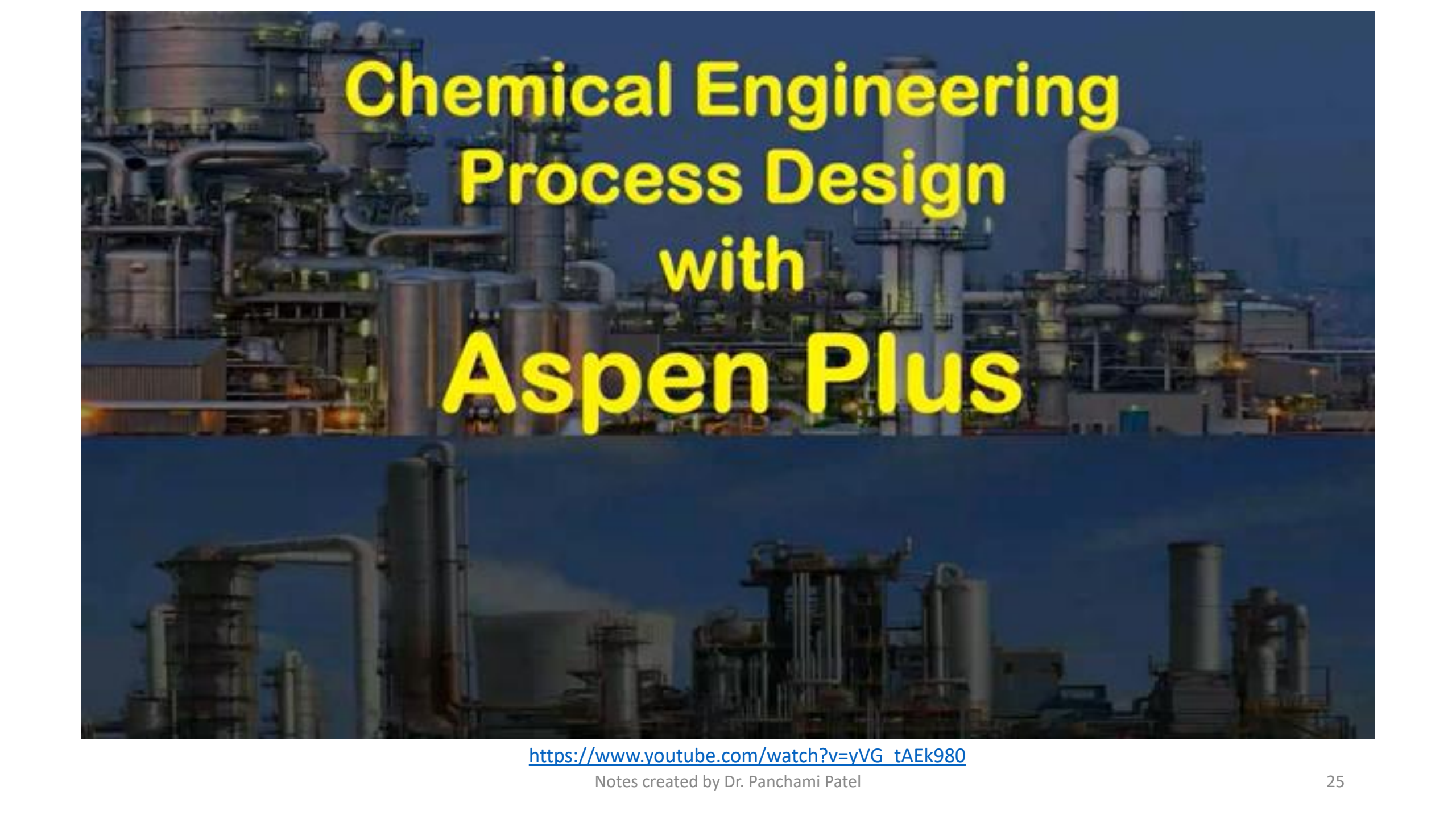
$$C_i = C_{if} + \tau r_i (C)$$

- The simplest kinetic reactor model is the **CSTR (continuous-stirred-tank reactor)**, in which **the contents are assumed to be perfectly mixed**.
- Thus, the **composition and the temperature are assumed to be uniform** throughout the reactor volume and equal to the composition and temperature of the reactor effluent.
- However, the fluid elements do not all have the same residence time in the reactor, there is a **residence-time distribution**.
- Can provide perfect mixing of the fluid contents of a vessel to approximate a CSTR model in a commercial reactor.
- A perfectly mixed reactor is used often for homogeneous liquid-phase reactions.
- The CSTR model is adequate for this case, provided that the reaction takes place under adiabatic or isothermal conditions.
- Although calculations only involve **algebraic equations, they may be nonlinear**, can have multiple solutions, two or more of which may be stable.

PFR

$$\frac{dc_i}{d\tau} = r_i (C)$$

- More complex is the **plug-flow tubular reactor (PFR or PFTR)**, in which the **composition of the fluid, flowing as a plug, gradually changes down the length of the reactor**, with no composition or temperature gradients in the radial direction.
- Furthermore, mass and heat transfer rates are negligible in the axial direction.
- Thus, **the PFR is completely unmixed**, with all fluid elements having the **same residence time in the reactor**.
- If the reactor operates under adiabatic or nonisothermal conditions, the temperature of the flowing fluid changes gradually down the length of the reactor.
- All simulators provide one-dimensional, plug-flow models that neglect axial dispersion.
- Thus, there are no radial gradients of temperature, composition, or pressure; and mass diffusion and heat conduction do not occur in the axial direction.
- Operation of the reactor can be adiabatic, isothermal, or nonadiabatic, nonisothermal.



Chemical Engineering Process Design with Aspen Plus

https://www.youtube.com/watch?v=yVG_tAEk980

Notes created by Dr. Panchami Patel

Reactor models – based on flow rates

Condition	Model
High flow rates (high Reynolds numbers) in a long tubular reactor	• PFR model • turbulent flow may approximate plug flow without appreciable axial mass and heat transfer
Reynolds numbers < 2100	• laminar flow persists • Partially mixed condition • PFR model not valid due to parabolic (nonplug) velocity profile • PFR model used with a 15% safety factor

Reactor models – based on phase of reactants

Condition	Model
Liquid-phase reactions	• Single PFR or CSTR reactor often used
Gas-phase reaction	• PFR preferred • CSTR has difficulty in obtaining perfect mixing in the gas
Noncatalytic homogeneous reactions	• Tubular reactor • can handle liquid or vapor feeds, with or without phase change in the reactor
Single reaction at isothermal conditions	• PFR • volume of PFR is smaller than CSTR for same conversion and temperature
Autocatalytic reactions (reaction rate depends on the concentration of a product) Autothermal reactions (feed is cold but reaction is highly exothermic)	• CSTR • volume of CSTR can be smaller than PFR, • such that axial dispersion in a tubular reactor may actually be beneficial

Reactor models – thermal models

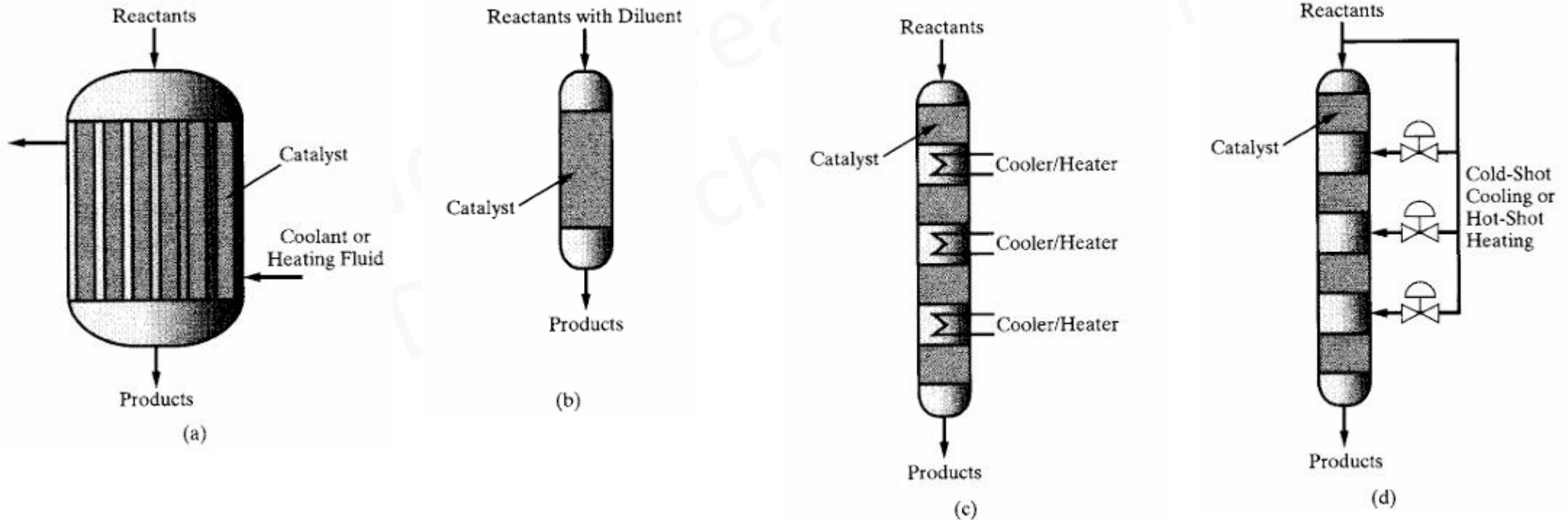
- For **fixed-bed catalytic reactors**, a **PFR model** with a pseudo-homogeneous kinetic equation is used and is a **1-D (one-dimensional) model**.
- However, if the **reactor is nonadiabatic with heat transfer** to or from the wall, the PFR model is not usually adequate and a **2-D model, involving the solution of partial differential equations for variations in temperature and composition in both the axial and radial directions**, is necessary.
- Simulators do not include 2-D models, but they can be generated by the user and inserted into the simulator.
- Models for fluidized-bed catalytic reactors are the most complex and cannot be adequately modeled with either the CSTR or PFR models.
- Because some of the gas passing through **the fluidized bed** can bypass the suspended catalyst, the **conversion in a fluidized bed can be less than that predicted by the CSTR model**.

Reactor design for complex configurations

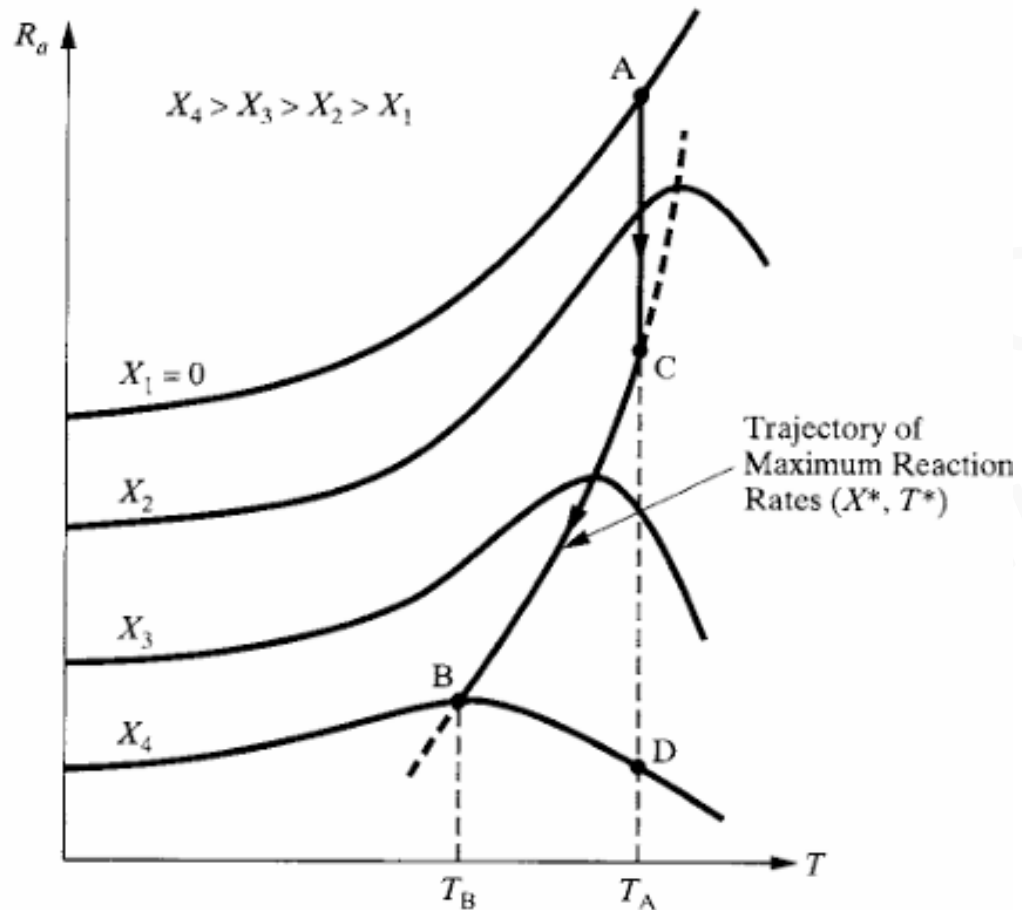
- **Temperature control is an important consideration in reactor design.**
- **Adiabatic operation** is always considered first because it provides the simplest and least-expensive reactor.
- Measure of the degree of exothermicity or endothermicity of a reaction is the **adiabatic temperature rise (ATR) for complete reaction** with reactants in stoichiometric ratio.
- **Methods to maintain adiabatic operation of endothermic/exothermic reactions:**
- (a) heat transfer to or from the reacting fluid, across a wall, to or from an **external cooling or heating agent** (e.g. manufacture of phthalic anhydride with V_2O_5 catalyst at 375 °C, ATR of 1170 °C, tubes with catalyst and reactant, sodium nitrite-potassium nitrate fused salt in shell for cooling)
- (b) an **inert or reactive heat carrier or diluent in the reacting fluid** (e.g. styrene from catalytic dehydrogenation of ethylbenzene at 575 °C, ATR of -460°C, inert steam at 625°C)
- (c) a **series of reactor beds with a heat exchanger for cooling or heating** between each pair of beds (e.g. Sulfur trioxide by catalytic oxidation of sulfur dioxide in air with V_2O_5 catalyst at 450 °C, ATR of 710 °C, reactor system consists of four adiabatic reactor beds with heat exchangers)

Reactor design for complex configurations

- (d) **cold-shot cooling** (also called direct-contact quench) or **hot-shot heating**, where the combined feed is split into two or more parts, one of which enters at the reactor entrance while the remaining parts enter the reactor at other locations. (e.g. ATR is higher, manufacture of ammonia from synthesis gas)

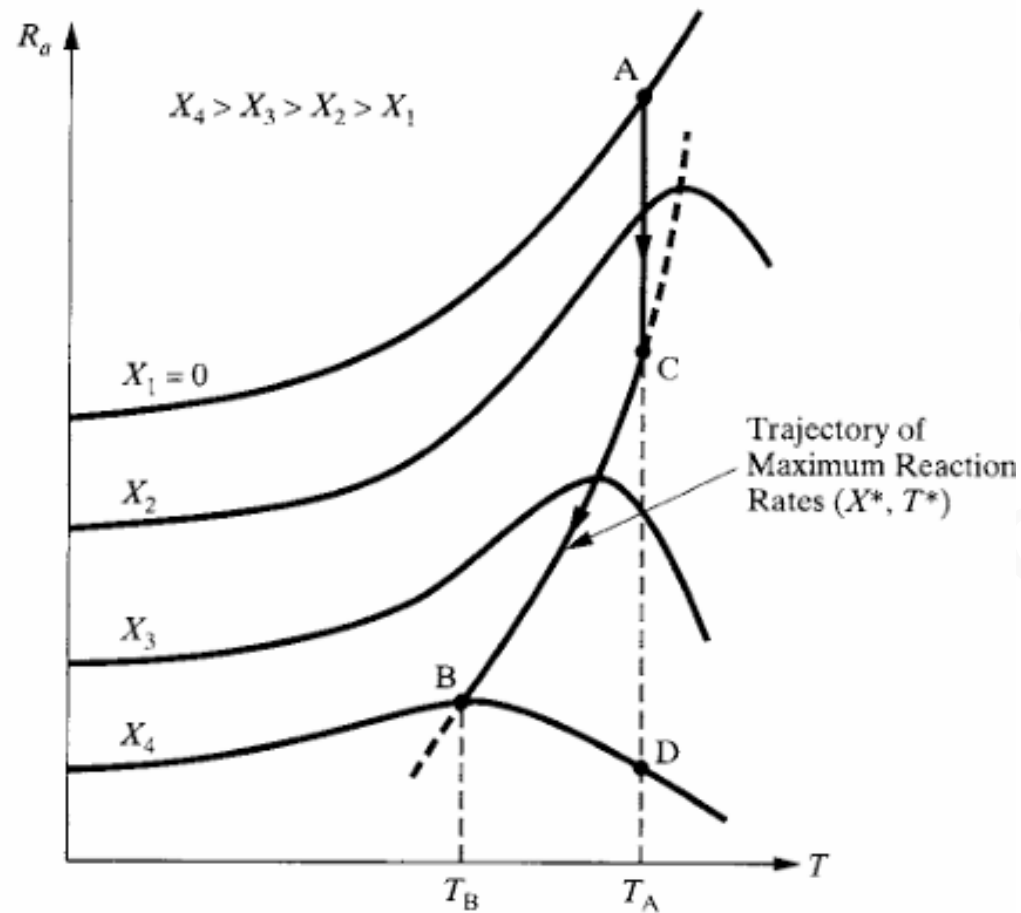


Reactor design for complex configurations



- For 1-D fixed-bed catalytic reactors, it is desirable to reduce the vessel volume to a minimum.
- Trajectory of mass- and energy-balance equations ($X(z)$, $T(z)$) is adjusted to match trajectory corresponding to maximum reaction rate (X^* , T^*)
- Reactors designed so that: $(X(z), T(z)) \approx (X^*, T^*)$.
- Specify: a temperature profile for a PFR or an exit temperature for a CSTR
- To determine: optimal reactor volume with required corresponding heat-duty profile (matched by heat exchange).

Reactor design for complex configurations



- For each value of X , the reaction rate curve shows a maximum value.
- **A locus of maximum rates corresponding to solid arc (dashed) passing through points C and B.**
- The maximum reaction rate decreasing with increasing fractional conversion.
- **At each conversion level, the desired temperature corresponds to the maximum reaction rate.**
- If entering temperature cannot be increased, best to operate isothermally at T_A until conversion at point C reached, and then follow optimal profile CB to the desired conversion.

Reactor design for complex configurations

Stability of an autothermal reactor

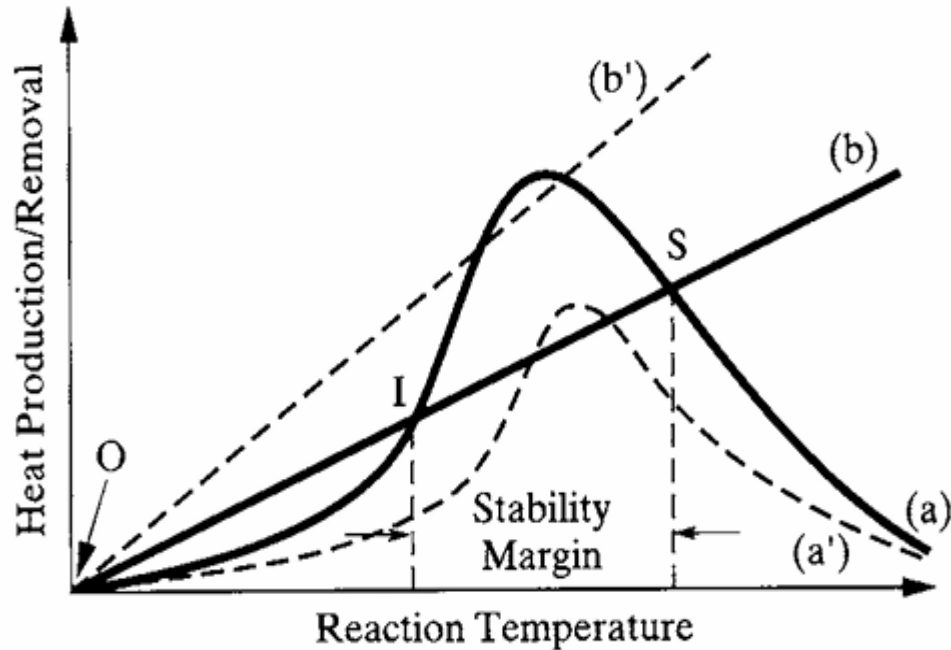


Figure 6.3 Multiple steady states in an autothermal reactor, with reaction rate limited by equilibrium: heat production rates for fully active (a) and deactivated catalyst (a'); heat removal rates for normal (b), and increased heat transfer (b').

- Autothermal reactor: a reactor whose feed is preheated by its effluent.
- For a reversible exothermic reaction, heat generation rate varies nonlinearly with reaction temperature.
- At low temperatures, the rate of heat generation is limited by the low rate of the forward reaction.
- At very high temperatures, the rate of reaction is limited by equilibrium, low heat generation rates are to be expected.
- **However, at some intermediate temperature, the reaction rate exhibits a maximum value.**

Reactor design for complex configurations

Stability of an autothermal reactor

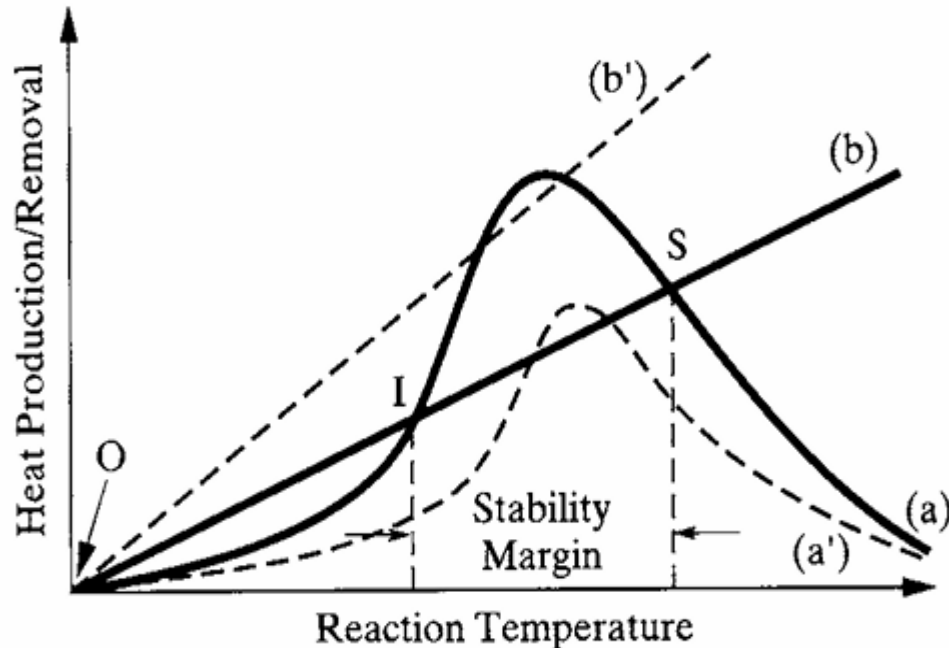


Figure 6.3 Multiple steady states in an autothermal reactor, with reaction rate limited by equilibrium: heat production rates for fully active (a) and deactivated catalyst (a'); heat removal rates for normal (b), and increased heat transfer (b').

- As heat transfer by convection is dominant, the rate of heat removal is almost linear with the reaction temperature
- With a slope dependent on the degree of heat exchange between the outlet and the inlet.
- Thus, the **intersection of the heat removal line (b) and the heat generation line (a)** sometimes leads to **three possible operating conditions**:
 - **O**: the non-reacting state (*stable*)
 - **I**: the ignition point (*unstable*)
 - **S**: the desired operating point (*stable*)

Reactor design for complex configurations

Stability of an autothermal reactor

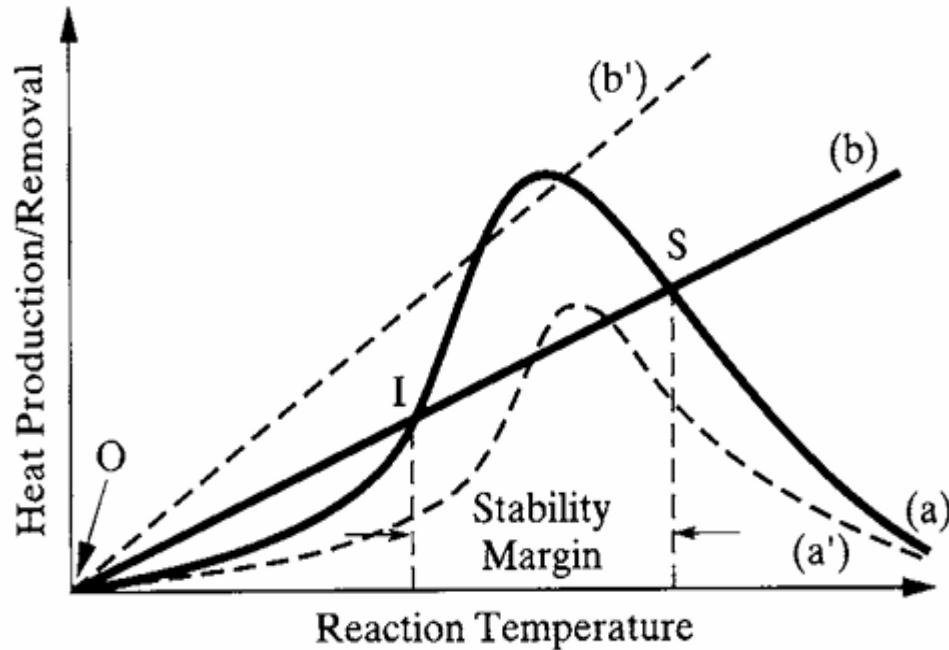


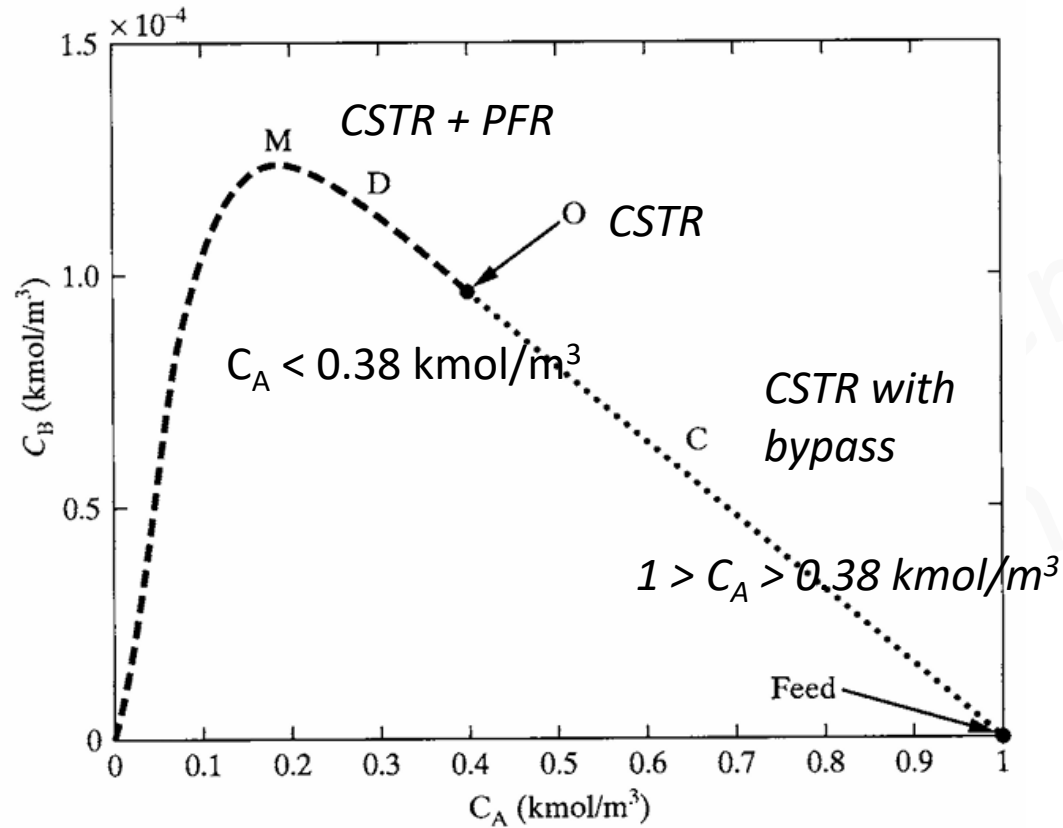
Figure 6.3 Multiple steady states in an autothermal reactor, with reaction rate limited by equilibrium: heat production rates for fully active (a) and deactivated catalyst (a'); heat removal rates for normal (b), and increased heat transfer (b').

- **Ignition point is *unstable*** because a small positive perturbation in temperature leads to a jump to the desired, stable operating point, whereas a negative perturbation leads to stable non-reacting state.
- **Temperature difference between operating points I and S as the "stability margin."**
- The proximity of the stable operating point to the unstable ignition point leads to an increased likelihood of **loss of control** in the face of process upsets.

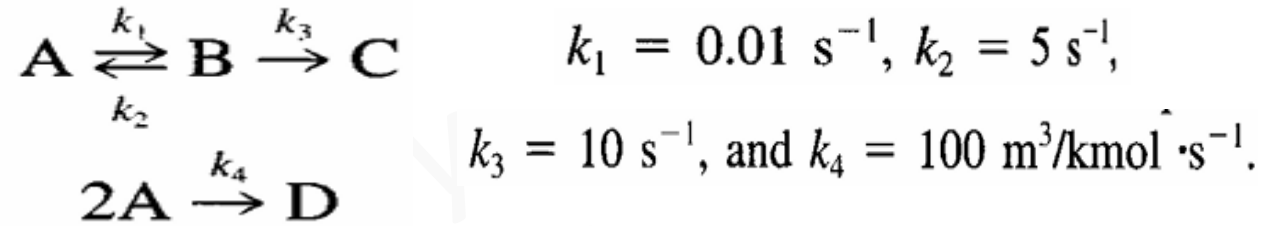
Reactor Network Design using the Attainable region

- **Attainable region (AR):** defines the achievable compositions that may be obtained from a network of chemical reactors.
- The attainable region in composition space was introduced by **Hom** (1964), with recent developments and extensions by **Glasser and co-workers** (Glasser et al. 1987; Hildebrandt et al., 1990).
- **The boundary of the attainable region is composed of arcs, each of which results from the application of a distinct reactor type.**

Reactor Network Design using the Attainable region



With a feed of 1 kmol/m³ of A



Attainable region for van de Vusse kinetics (van de Vusse, 1964)

- Within the region bounded by the three arcs and the horizontal base line ($C_B = 0$), product compositions can be achieved with some combination of these reactor configurations.
- Maximum achievable concentration, $C_B = 1.25 \times 10^{-4}$ kmol/m³, is obtained using a CSTR followed by a PFR (at point M along curve D).

AR for CSTR

- Finding the CSTR effluent concentration C requires the solution of a system of nonlinear equations.

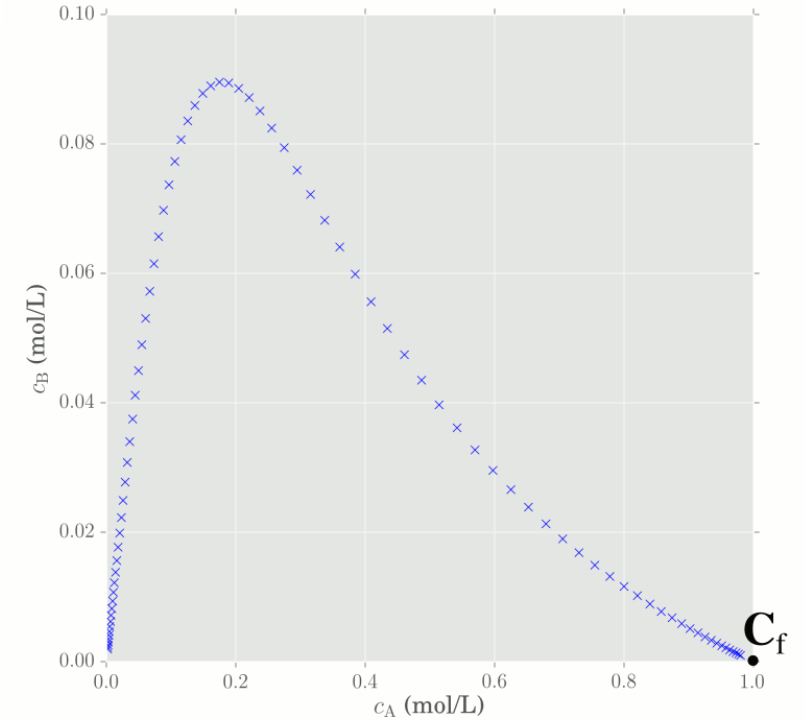
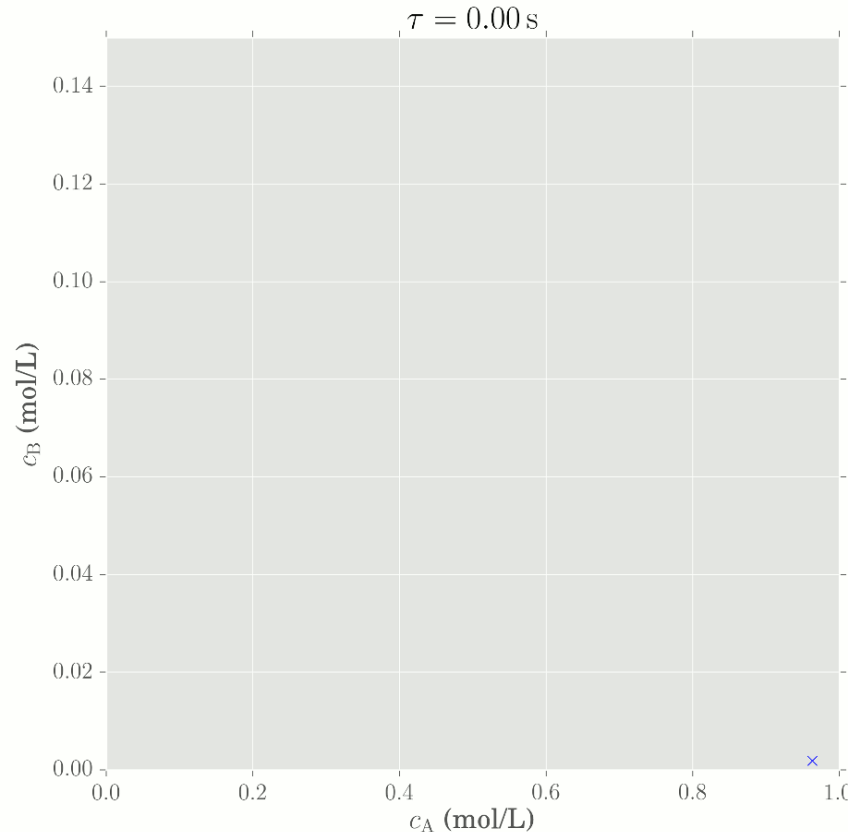
$$c_i = c_{if} + \tau r_i(C)$$

$$\begin{bmatrix} c_A \\ c_B \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} c_{Af} \\ c_{Bf} \\ \vdots \\ c_{nf} \end{bmatrix} + \tau \begin{bmatrix} r_A(C) \\ r_B(C) \\ \vdots \\ r_n(C) \end{bmatrix}$$

$$C = C_f + \tau r(C)$$

$$(C - C_f) = \tau r(C)$$

mixing vector $(C - C_f)$ is a scalar multiple of vector $r(C)$



rate vector evaluated at a CSTR effluent concentration C , operated with a CSTR feed concentration of C_f is colinear to the CSTR mixing vector $(C - C_f)$

https://attainableregions.com/examples/ar_fundamentals.html

AR for PFR

- System of ordinary differential equations (ODEs), solution an initial condition (feed concentration) and integration time (PFR residence time).
- Solving the system of ODEs results in a **PFR solution trajectory**.

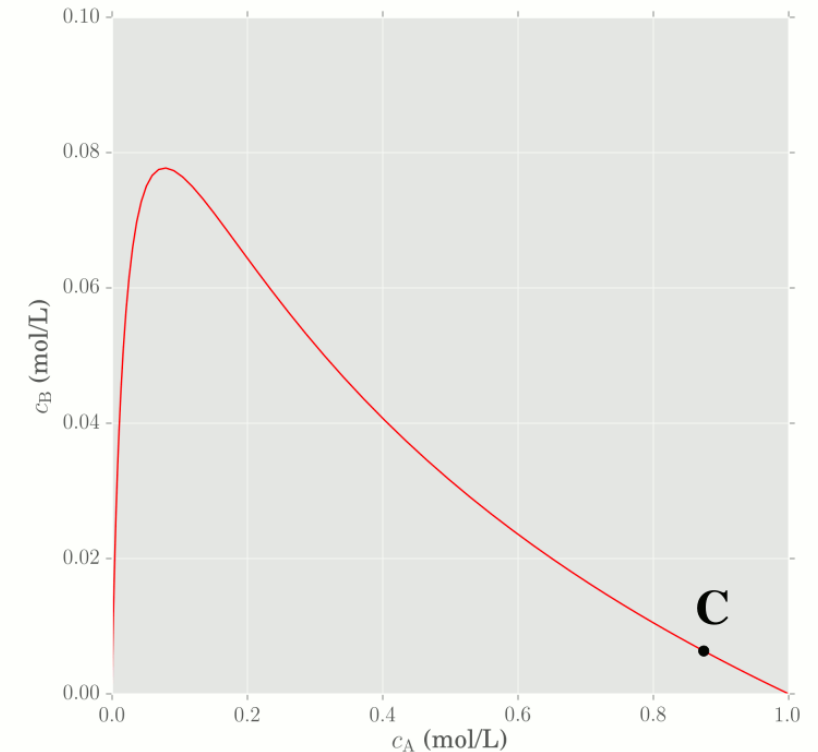
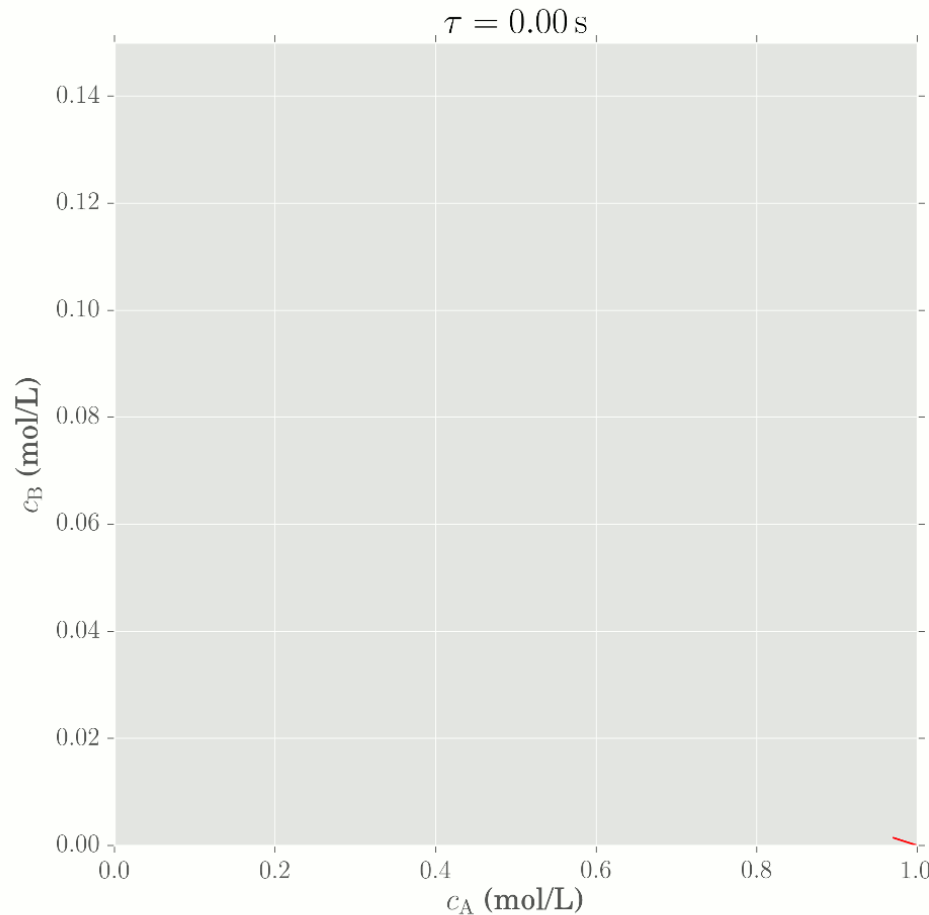
$$\frac{dc_i}{d\tau} = r_i(\mathbf{C})$$

$$\begin{bmatrix} \frac{dc_A}{d\tau} \\ \frac{dc_B}{d\tau} \\ \vdots \\ \frac{dc_n}{d\tau} \end{bmatrix} = \begin{bmatrix} r_A(\mathbf{C}) \\ r_B(\mathbf{C}) \\ \vdots \\ r_n(\mathbf{C}) \end{bmatrix}$$

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{r}(\mathbf{C})$$

Gradient of
PFR trajectory
with respect to
PFR residence
time evaluated
at C

Rate vector
evaluated
at C



Rate vector evaluated at any point along a
PFR solution trajectory is tangent to the
trajectory

https://attainableregions.com/examples/ar_fundamentals.html

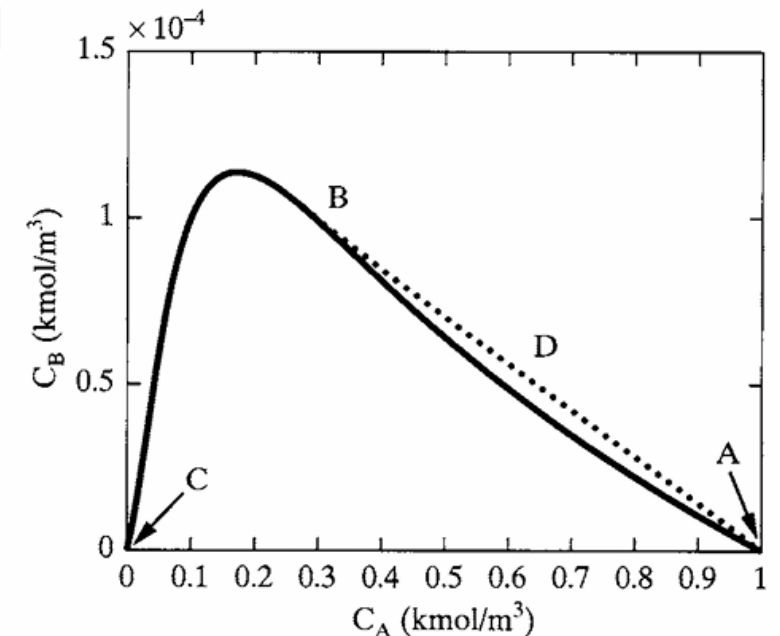
Construction of the Attainable region

Step 1:

- Begin by **constructing a trajectory for a PFR from the feed point, continuing to the complete conversion of A or chemical equilibrium.**
- The PFR trajectory is computed by solving simultaneously the kinetic equations for A and B, where τ is the PFR residence time.
- The **trajectory in $C_A - C_B$ space** is plotted as curve ABC for case that Component A is completely converted.

$$\frac{dC_A}{d\tau} = -k_1 C_A + k_2 C_B - k_4 C_A^2$$

$$\frac{dC_B}{d\tau} = k_1 C_A - k_2 C_B - k_3 C_B$$



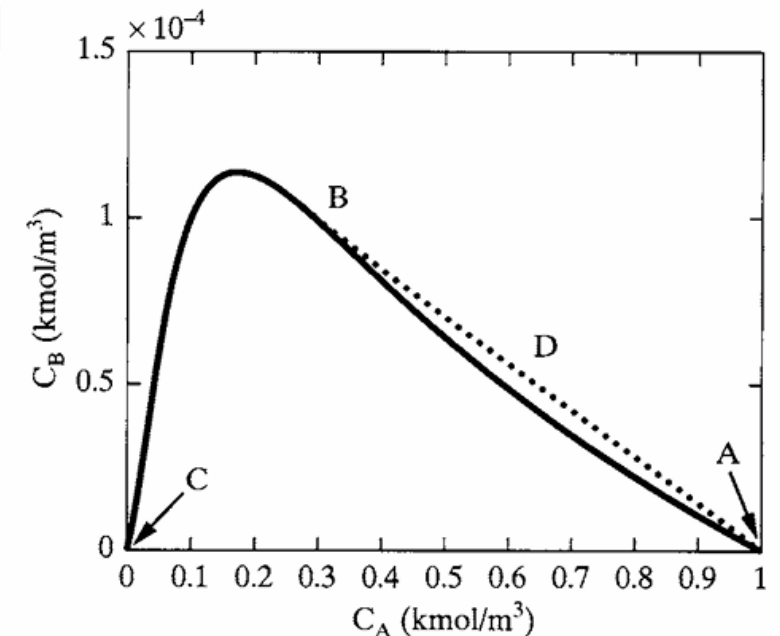
Construction of the Attainable region

Step 2:

- When the PFR trajectory bounds a convex region, this constitutes a candidate attainable region.
- A convex region is one in which all straight lines drawn from one point on the boundary to any other point on the boundary lie wholly within the region or on the boundary.
- When the rate vectors $[dC_A/d\tau, dC_B/d\tau]^T$ at concentrations outside of the candidate AR do not point back into it, the current limits are the boundary of the AR and the procedure terminates.
- Here the PFR trajectory is not convex from A to B, so proceed to the next step to determine if the attainable region can be extended beyond the curve ABC.

$$\frac{dC_A}{d\tau} = -k_1 C_A + k_2 C_B - k_4 C_A^2$$

$$\frac{dC_B}{d\tau} = k_1 C_A - k_2 C_B - k_3 C_B$$



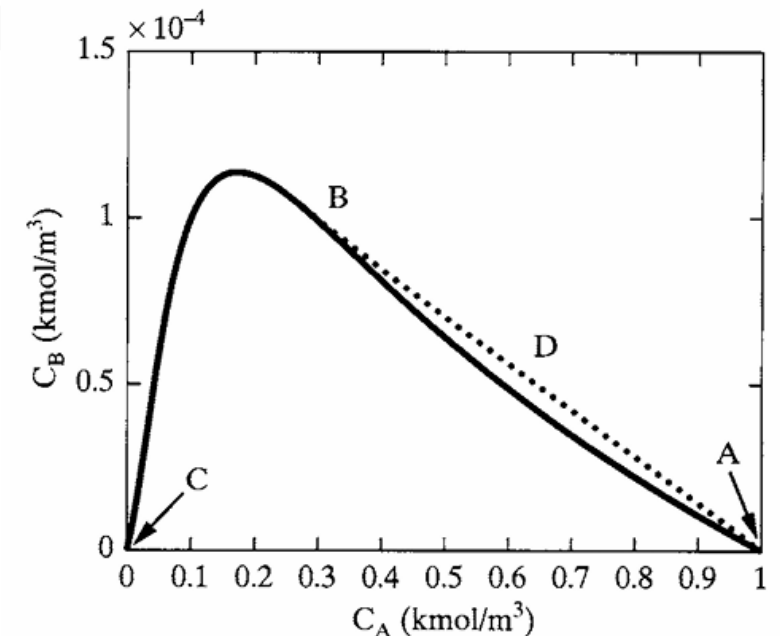
Construction of the Attainable region

Step 3:

- The **PFR trajectory is expanded by linear arcs**, representing mixing between the PFR effluent and the feed stream, extending the candidate attainable region.
- A linear arc, ADB, is added, extending the attainable region to ADBC.
- Rate vectors computed along this arc are found to point out of the extended AR, proceed to the next step.

$$\frac{dC_A}{d\tau} = -k_1 C_A + k_2 C_B - k_4 C_A^2$$

$$\frac{dC_B}{d\tau} = k_1 C_A - k_2 C_B - k_3 C_B$$



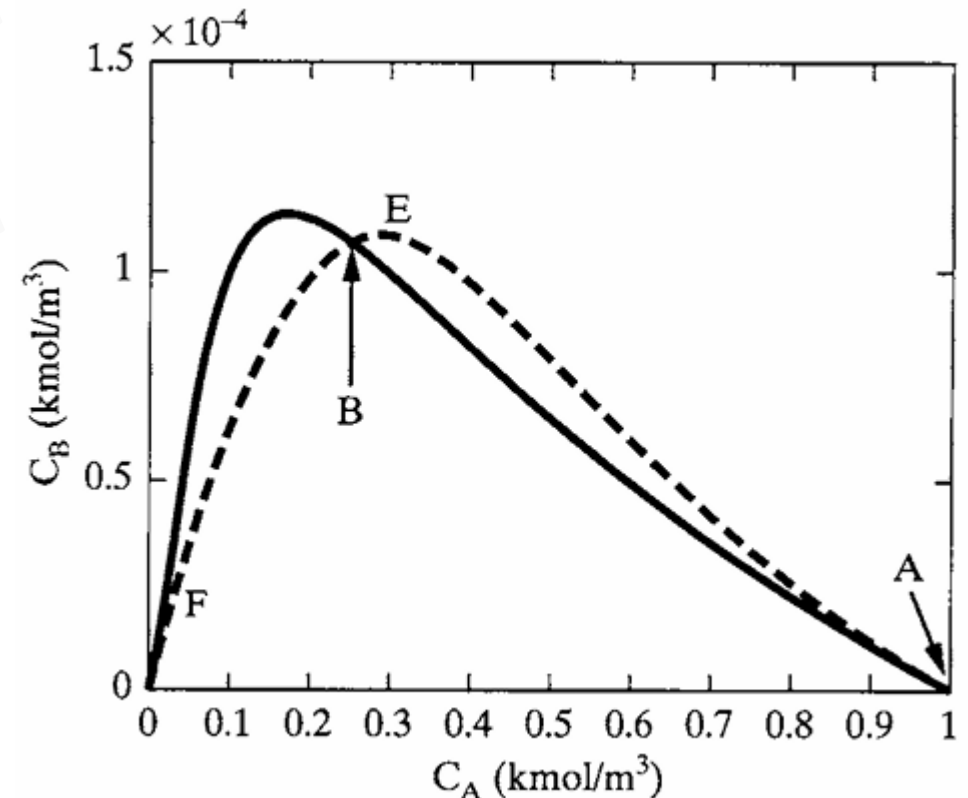
Construction of the Attainable region

Step 4:

- Possible that **CSTR trajectory** enlarges attainable region.
- After **placing CSTR trajectory that extends AR most**, additional linear arcs that represent mixing of streams are placed to **ensure that AR remains convex**.
- The CSTR trajectory is computed by solving the CSTR form of the kinetic equations for A and B.
- The CSTR trajectory that extends the AR most is that computed from feed point, at C_{A0} , largest concentration of A: curve AEF which passes through point B.
- Since the union of the previous AR and the CSTR trajectory is not convex, a linear arc, **AGO**, is **augmented**.
- **This arc represents a CSTR with a bypass stream.**

$$C_{A0} - C_A = \tau(k_1 C_A - k_2 C_B + k_4 C_A^2)$$

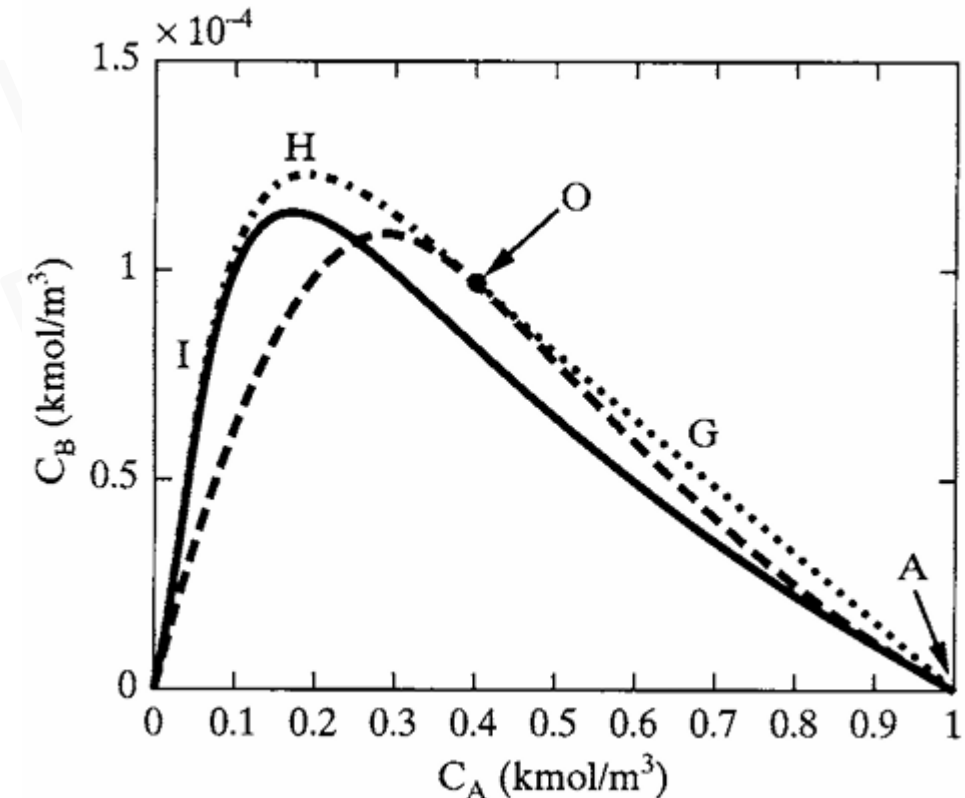
$$C_B = \tau(k_1 C_A - k_2 C_B - k_3 C_B)$$



Construction of the Attainable region

Step 5:

- A PFR trajectory is drawn from the position where the mixing line meets the CSTR trajectory.
- If this PFR trajectory is convex, it extends the previous AR to form an expanded candidate AR. Then return to Step 2. Otherwise, repeat the procedure from Step 3.
- The PFR trajectory, OHI leads to a convex attainable region.
- The boundaries of region are: (a) the linear arc, AGO (CSTR with bypass stream) (b) the point O (CSTR) and (c) the arc OHI (CSTR followed by a PFR in series).
- It is noted that the maximum composition of B is obtained at point H, using a CSTR followed by a PFR.



Quick Recap

- What are the type of reactor models?
- What is conversion and extent of reaction?
- What is the equation for the CSTR model?
- What is the equation for the PFR model?
- What are some methods to maintain adiabatic operation in a reactor?
- What are the possible operating conditions for stability of an autothermal reactor?
- What is attainable region for reaction?